

Relaxed Randomized Gossip on the Complete Graph: Exact Covariance Spectrum and Sharp Consensus Law

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3 September 2026

Abstract

For uniform pairwise relaxed averaging on the complete graph, we derive the exact first moment and sharp mean-square disagreement law at every finite time. The entire second-moment transfer on $\text{Sym}^2(\mathbf{1}^\perp)$ splits into three explicit orthogonal blocks; all eigenvalues, multiplicities, projectors, and matrix-valued iterates are closed.

1 Frozen stochastic owner

Fix $N \geq 2$ and $0 \leq \eta \leq 1$. Independently at every integer time, choose an unordered pair $\{i, j\}$ uniformly from the edges of K_N and set

$$x_{t+1} = W_{ij}x_t, \quad W_{ij} = I - \eta d_{ij}d_{ij}^\top, \quad d_{ij} = e_i - e_j. \quad (1)$$

Thus the selected entries become $(1 - \eta)x_i + \eta x_j$ and $\eta x_i + (1 - \eta)x_j$. The endpoint $\eta = 1/2$ is ordinary pairwise averaging, while $\eta = 1$ merely swaps the two entries. The initial vector is deterministic.

Put

$$P = I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top, \quad \bar{x} = \frac{1}{N}\mathbf{1}^\top x_0, \quad y_t = Px_t = x_t - \bar{x}\mathbf{1}.$$

Every W_{ij} is symmetric, fixes $\mathbf{1}$, and commutes with P . The basic complete-graph identity is

$$\sum_{i < j} d_{ij}d_{ij}^\top = NP. \quad (2)$$

Theorem 1 (Mean and sharp energy law). *The sample mean is invariant on every realization. With*

$$\mu = 1 - \frac{2\eta}{N-1}, \quad \rho = 1 - \frac{4\eta(1-\eta)}{N-1}, \quad (3)$$

one has, for every $t \geq 0$,

$$\mathbb{E}y_t = \mu^t y_0, \quad \mathbb{E}\|y_t\|^2 = \rho^t \|y_0\|^2. \quad (4)$$

Proof. Equation (1) fixes $\mathbf{1}^\top x$ pathwise. Since K_N has $N(N-1)/2$ edges, averaging (1) and using (2) gives

$$\mathbb{E}W = I - \frac{2\eta}{N-1}P,$$

which acts by μ on $\mathbf{1}^\perp$. Independence of the new edge from the past proves the first identity by iteration. Since $(d_{ij}d_{ij}^\top)^2 = 2d_{ij}d_{ij}^\top$,

$$W_{ij}^2 = I - 2\eta(1-\eta)d_{ij}d_{ij}^\top.$$

Conditioning on y_t , applying (2), and iterating gives the second identity. It is an equality for every initial vector, hence its rate is sharp whenever $y_0 \neq 0$. $\square$

The phrase *sharp energy owner* records the content added in the original paper round: no spectral conclusion is needed for Theorem 1.

2 Three invariant covariance blocks

Let

$$\mathcal{H}_N = \{A \in \mathbb{R}^{N \times N} : A = A^\top, PAP = A\}$$

with Frobenius inner product. For $N \geq 3$ define

$$\begin{aligned}\mathcal{V}_0 &= \text{span}\{P\}, \\ \mathcal{V}_1 &= \{P \text{diag}(u)P : \mathbf{1}^\top u = 0\}, \\ \mathcal{V}_2 &= \{A \in \mathcal{H}_N : \text{diag } A = 0\}.\end{aligned}$$

For $N = 3$, $\mathcal{V}_2 = \{0\}$. For $N = 2$, only $\mathcal{V}_0 = \mathcal{H}_2$ remains.

Lemma 2 (Explicit orthogonal projectors). *For $N \geq 3$ and $A \in \mathcal{H}_N$, set*

$$\begin{aligned}\Pi_0 A &= \frac{\text{tr } A}{N-1} P, & R &= A - \Pi_0 A, \\ u &= \frac{N}{N-2} \text{diag } R, & \Pi_1 A &= P \text{diag}(u)P, & \Pi_2 A &= A - \Pi_0 A - \Pi_1 A.\end{aligned}\tag{5}$$

Then Π_r are the orthogonal projectors onto $\mathcal{V}_r$, and

$$\dim \mathcal{V}_0 = 1, \quad \dim \mathcal{V}_1 = N-1, \quad \dim \mathcal{V}_2 = \frac{N(N-3)}{2}.\tag{6}$$

Proof. The first residual has trace zero, so $\mathbf{1}^\top u = 0$. For such u ,

$$\text{diag}(P \text{diag}(u)P) = \frac{N-2}{N} u.$$

Hence $\Pi_1 A$ has the same diagonal as R , and $\Pi_2 A$ has zero diagonal. Centered matrices with zero diagonal are Frobenius orthogonal to both P and every centered diagonal pattern. This proves orthogonality and the projector formulas. The map $u \mapsto P \text{diag}(u)P$ is injective on $\mathbf{1}^\perp$ for $N > 2$. Subtracting dimensions from $\dim \mathcal{H}_N = N(N-1)/2$ proves (6). $\square$

Define the second-moment transfer

$$\mathcal{T}_\eta(A) = \frac{2}{N(N-1)} \sum_{i < j} W_{ij} A W_{ij}.\tag{7}$$

Theorem 3 (Complete second-moment spectrum). *The three spaces in Lemma 2 are orthogonal invariant blocks of $\mathcal{T}_\eta$ for every $0 \leq \eta \leq 1$. The restrictions to the present nonzero blocks are scalar, with respective values*

$$\lambda_0 = 1 - \frac{4\eta(1-\eta)}{N-1},\tag{8}$$

$$\lambda_1 = 1 - \frac{4\eta - 2\eta^2}{N-1},\tag{9}$$

$$\lambda_2 = 1 - \frac{4\eta}{N-1} + \frac{4\eta^2}{N(N-1)}.\tag{10}$$

Absent low-dimensional blocks are omitted. If $\eta > 0$, the values attached to the present blocks are pairwise distinct, so those blocks are precisely the corresponding full eigenspaces. If $\eta = 0$, then $\mathcal{T}_0 = I$: the only eigenvalue is one, its eigenspace is all of $\mathcal{H}_N$, and its multiplicity is $N(N-1)/2$.

Proof. For centered A , expansion of (7) gives

$$\mathcal{T}_\eta(A) = \left(1 - \frac{4\eta}{N-1}\right)A + \frac{2\eta^2}{N(N-1)}\mathcal{S}(A), \quad \mathcal{S}(A) = \sum_{i < j} (d_{ij}^\top A d_{ij}) d_{ij} d_{ij}^\top.\tag{11}$$

If $A = P$, every scalar coefficient in $\mathcal{S}$ is two, so $\mathcal{S}(P) = 2NP$. If $A = P \text{diag}(u)P$ with $\mathbf{1}^\top u = 0$, then $d_{ij}^\top A d_{ij} = u_i + u_j$. The off-diagonal and diagonal entries give $\mathcal{S}(A) = NA$. Finally, for $A \in \mathcal{V}_2$, $d_{ij}^\top A d_{ij} = -2a_{ij}$; its zero row sums then give $\mathcal{S}(A) = 2A$. Substitution in (11) yields the three displayed eigenvalues. For $\eta > 0$,

$$\lambda_0 - \lambda_1 = \frac{2\eta^2}{N-1} > 0, \quad \lambda_1 - \lambda_2 = \frac{2\eta^2(N-2)}{N(N-1)} > 0,$$

where the second comparison is used only when $\mathcal{V}_2$ is present. Together with the orthogonal decomposition, this proves the full-eigenspace statement. At $\eta = 0$ every $W_{ij} = I$, which proves the asserted merger. $\square$

Theorem 4 (Exact second moment). *Let $A_0 = y_0 y_0^\top$ and $M_t = \mathbb{E}[y_t y_t^\top]$. For $N \geq 3$,*

$$M_t = \lambda_0^t \Pi_0 A_0 + \lambda_1^t \Pi_1 A_0 + \lambda_2^t \Pi_2 A_0, \quad (12)$$

where the last term is absent at $N = 3$. At $N = 2$, $M_t = \lambda_0^t A_0$.

Proof. The newly selected edge is independent of the past, hence $M_{t+1} = \mathcal{T}_\eta(M_t)$. Apply Theorem 3 and iterate. Taking traces recovers Theorem 1, because the two trace-free blocks do not contribute. $\square$

References

- [1] S. Boyd, A. Ghosh, B. Prabhakar, and D. Shah, “Randomized gossip algorithms,” *IEEE Transactions on Information Theory* 52 (2006), 2508–2530. DOI: [10.1109/TIT.2006.874516](https://doi.org/10.1109/TIT.2006.874516).

The source establishes the randomized-gossip owner and network setting. The complete-graph formulas above are derived directly for the frozen model; no literature-priority claim is made for them.